Barcelona School of Economics

Master's Degree in Specialized Economic Analysis
International Trade, Finance, and Development

"The Anatomy of Structural Transformation: Disaggregating Services in African Economies"

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ABSTRACT IN ENGLISH:

Across Africa, labour is leaving agriculture without passing through manufacturing, flowing instead into consumer services. The key question is whether this expansion reflects productivity growth within consumer services or whether the sector is primarily absorbing workers released from agriculture, and how these different dynamics shape structural transformation. Extending Duarte and Restuccia (2010) to four sectors, our study separates consumer services from the broader service aggregate and applies counterfactual analysis to thirteen African economies over 1990–2018. We find substantial heterogeneity: consumer service productivity growth is the dominant driver of agricultural exit in some economies but suppresses it in others. It follows that the developmental significance of consumer service expansion depends on its productivity dynamics, and not just its scale.

ABSTRACT IN CATALAN:

Arreu d'Àfrica, la mà d'obra abandona l'agricultura sense passar per la indústria manufacturera, i flueix en canvi cap als serveis de consum. La qüestió clau és si aquesta expansió reflecteix un creixement de la productivitat dins dels serveis de consum o si el sector està absorbint principalment treballadors alliberats de l'agricultura, i com aquestes dinàmiques diferents configuren la transformació estructural. Ampliant el model de Duarte i Restuccia (2010) a quatre sectors, el nostre estudi separa els serveis de consum de l'agregat de serveis més ampli i aplica una anàlisi contrafactual a tretze economies africanes durant el període 1990–2018. Trobem una heterogeneïtat substancial: el creixement de la productivitat dels serveis de consum és el principal motor de la sortida de l'agricultura en algunes economies, però la frena en d'altres. Se'n desprèn que la rellevància per al desenvolupament de l'expansió dels serveis de consum depèn de la seva dinàmica de productivitat, i no només de la seva escala.

KEYWORDS (ENGLISH): STRUCTURAL TRANSFORMATION, CONSUMER SERVICES, PRODUCTIVITY

KEYWORDS (CATALAN): TRANSFORMACIÓ ESTRUCTURAL, SERVEIS DE CONSUM, PRODUCTIVITAT

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1 Introduction

“I have become skeptical about the viability of the traditional industrialization-led growth model,” Dani Rodrik (2026) recently reflected, arguing instead that “the evidence suggests the possibility of a virtuous cycle of economic growth built on middle-class services.” A decade ago, such a statement would have been surprising coming from Rodrik, one of the field’s most influential voices on economic development. Yet today, many developing economies are industrializing weakly or not at all, while employment growth increasingly occurs in services. The conventional pattern of structural transformation—where labour moves out of agriculture and into increasingly productive sectors, with manufacturing playing the central role—is no longer unfolding in its canonical form. The debate has shifted from whether manufacturing remains the primary pathway to development to whether services can generate the productivity growth and labour absorption once associated with industrialization.

The African continent provides a particularly relevant setting in which to examine this question. Across much of the region, labour has continued to leave agriculture despite manufacturing remaining small and often stagnant. Figure 1 shows that this labour is flowing directly into services rather than manufacturing across a subset of African countries. When the sector is disaggregated, much of this labour absorption is concentrated in consumer services such as retail trade, transportation, restaurants, and personal services (Appendix Figure 8). The central challenge is no longer whether structural transformation can occur without industrialization, but whether these expanding service sectors are generating productivity growth or simply absorbing labour released from agriculture.

Whether this pattern reflects a weaker form of structural transformation or the emergence of a new developmental pathway is therefore unclear. Traditional industrialization has become increasingly difficult to pursue, owing to factors such as China’s dominance of low-wage manufacturing, rising capital intensity, and shifting trade patterns (Rodrik, 2016; Diao et al., 2025). If consumer service expansion is driven by rising productivity within the sector, Africa may instead be charting a viable alternative route to structural transformation. The idea that services can act as an engine of economic development has gained prominence in recent years, with India emerging as the leading example of service-led structural transformation. Whether Africa is following a similar trajectory remains an open question.

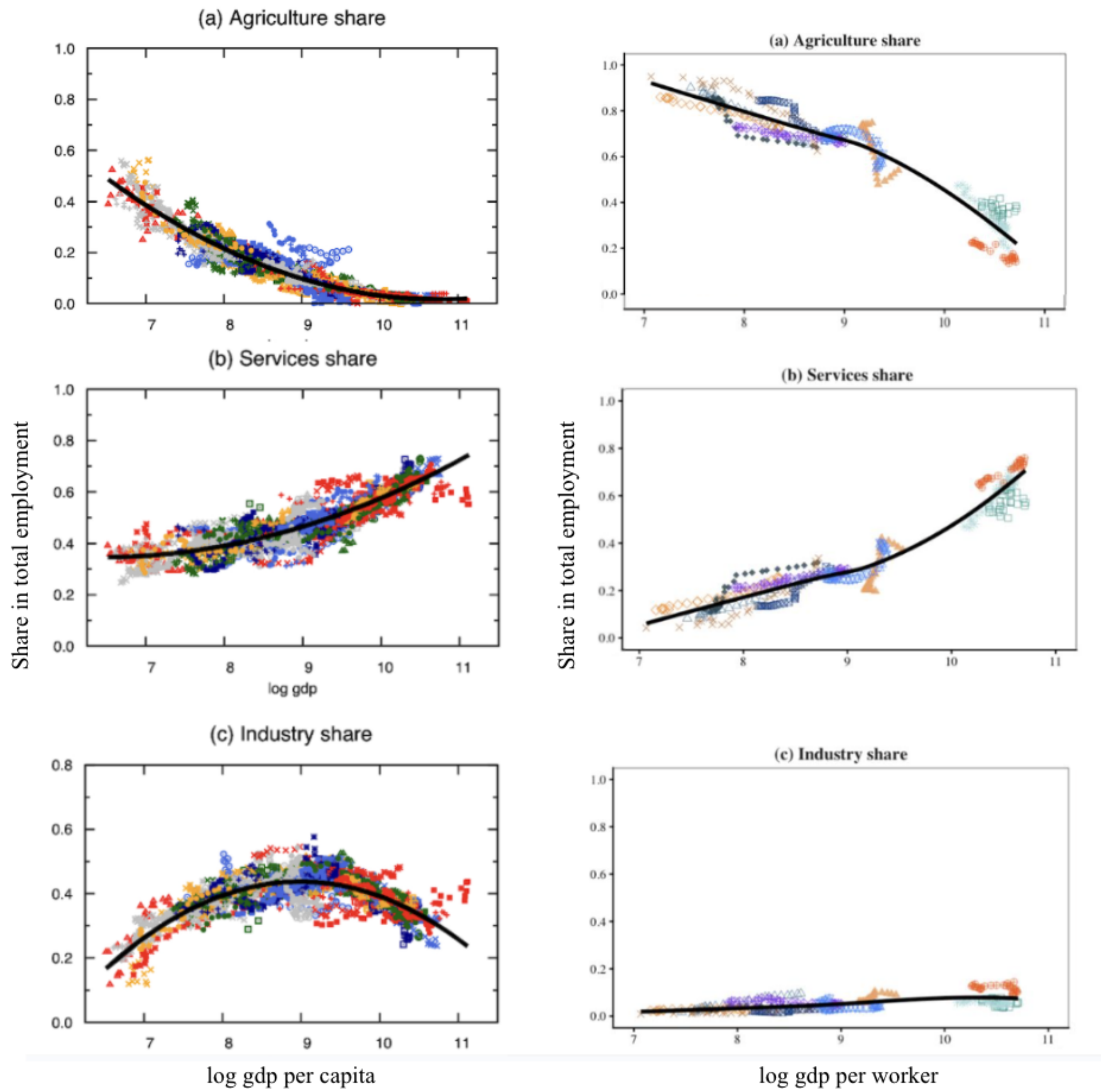

Figure 1: Sectoral Labour Shares against GDP per Capita/Worker
Source: García-Santana et al. (2021), PWT, UNU-WIDER ETD. Left panel: Evidence from a large international panel. Right panel: Authors' reproduction using African economies and reporting against log GDP per worker.

Existing frameworks are poorly equipped to answer this question because they treat services as a single aggregate sector. We address this limitation by analysing thirteen African economies alongside India as a point of comparison, drawing on sectoral employment and value added data from the UNU-WIDER Economic Transformation Database over 1990–2018. To do so, we extend the three-sector model of Duarte and Restuccia (2010) to four sectors, separating consumer services from the broader service aggregate to isolate its role in labour reallocation. Counterfactual analysis then decomposes what we observe into productivity-pull and income-push components at the consumer-service level, a decomposition not previously applied to a broad African sample.

Our results highlight that manufacturing employment shares have remained largely flat across the period, playing a limited role in structural transformation. Beyond this common pattern, there is substantial heterogeneity in the forces shaping structural transformation. Where consumer services productivity has grown, it emerges as a major contributor to agricultural exit, in some cases the dominant one, operating primarily through income effects. Conversely, where consumer services productivity has stagnated or declined, the sector acts as a drag on transformation, keeping labour in agriculture. Had these countries instead experienced India's consumer services productivity trajectory, agricultural employment shares would have been up to 14 percentage points lower by 2018.

These findings carry implications for both policy and method. In particular, they suggest that policymakers should distinguish between the expansion of service-sector employment and productivity growth within services, as the two do not necessarily coincide. More broadly, they underscore the limits of applying models built for early industrializers to low-income contexts, and suggest that disaggregating services is key to understanding structural transformation in these settings.

Our paper begins with a brief literature review in Section 2. Section 3 describes our data sources. Section 4 presents stylized facts for our country sample. Section 5 sets out our theoretical framework and calibration. Section 6 presents results and counterfactual analysis. Section 7 discusses implications and limitations, and Section 8 concludes.

2 Literature Review

Our paper builds on three strands of research: classical structural transformation theory, premature deindustrialization, and service-led growth in low-income economies. Lewis (1954) and Kuznets (1971) established the foundations of structural transformation theory, documenting the historical tendency for labour to move out of agriculture, through industry, and eventually into services as economies develop. This stylized pattern became the canonical account of economic development, with manufacturing occupying a central role in generating productivity growth and enabling catch-up with richer economies.[1]

[1] Early growth models such as Ramsey (1928) and Solow (1956) did not account for sectoral composition, treating differences between rich and poor countries as reflecting capital accumulation rather than structural differences in the economy.

Theoretical explanations of structural transformation typically emphasize two mechanisms. The first operates through income effects: as incomes rise, demand shifts away from food and toward manufactures and services, a pattern formalized by Kongsamut et al. (2001). In Alvarez-Cuadrado and Poschke (2011) terminology, rising agricultural productivity pushes workers off the land while demand shifts pull resources into industry and services. Boppart (2014) and Comin et al. (2021) show that these income effects are larger and more persistent than earlier models suggested. The second mechanism operates through substitution effects driven by relative prices. Baumol (1967) emphasized uneven productivity growth across sectors, while Ngai and Pissarides (2007) show that differential sectoral productivity growth can shift expenditure and labour toward services even when service-sector productivity grows more slowly.

The quantitative workhorse for studying these interactions is Duarte and Restuccia (2010), who develop a three-sector general equilibrium model in which structural transformation is driven by differential productivity growth and non-homothetic demand. The latter is commonly captured through Stone-Geary preferences (Geary, 1950; Stone, 1954), which generate the Engel curve dynamics underlying the income effect.[2] Herrendorf et al. (2014) document a broad consensus that non-homothetic preferences and differential productivity growth jointly account for the major Kuznets facts.

Despite its breadth, this literature was largely designed to explain the experience of advanced economies. Sen (2023) applies the Duarte-Restuccia framework to low-income economies and finds systematic prediction errors, concluding that this prototype neoclassical model can only explain structural transformation in rich countries. This finding directly motivates the framework we develop in this paper.

From the 2000s onward, a distinct subset of literature argued that late developers were not following the classical path, with labour flowing directly from agriculture into services rather than through a sustained manufacturing phase. McMillan and Rodrik (2014) document this pattern for Africa, showing that structural change reallocated labour toward below-average productivity sectors, as revealed by shift-share decomposition analysis. Rodrik (2016) termed this phenomenon *premature deindustrialization*, noting that developing countries now peak at 13 to 15 percent manufacturing employment, well below the levels achieved by early industrializers. Subsequent work shows that the manufacturing-income hump has shifted downward, jobless industrialization is widespread, and unconditional convergence in manufacturing productivity cannot be taken for granted (Sposi et al., 2025; Diao et al., 2025;

[2] Comin et al. (2021) propose a more flexible non-homothetic CES specification that accommodates richer patterns of income elasticity across sectors, extending the framework to later stages of development.

Herrendorf and Valentinyi, forthcoming; Bustos et al., 2016). Taken together, this literature suggests that African countries are following a qualitatively different structural path, raising the question of what drives labour absorption in manufacturing's absence.

A growing literature has asked whether consumer services can themselves drive structural transformation rather than merely reflect income growth elsewhere in the economy. This question is motivated by the heterogeneity of the service sector and the difficulty of measuring productivity in non-tradable consumer services, which standard measures classically understate (Lagakos, 2016). Fan et al. (2023) develop a structural model to infer consumer service productivity growth from observed employment shifts, finding that consumer service productivity was a first-order driver of structural transformation in India between 1987 and 2011. Peters et al. (2026) extend this approach to Africa, finding that consumer services productivity accounts for a meaningful share of structural transformation gains, though data limitations restrict their sample to six countries.

More research is needed to establish whether these findings generalise across the African continent, and through what mechanisms consumer service productivity shapes the broader process of structural transformation. Our paper addresses both questions: We extend the Duarte-Restuccia framework to four sectors, separating consumer services from the broader service aggregate, and apply it to a range of African countries and India. This allows us to decompose structural transformation into productivity-pull and income-push components at the consumer-service level for a broader African sample than previously studied.

3 Data Sources

This section outlines the data sources underlying our calibration exercise and stylized facts, and describes the construction of key variables. We use data from thirteen geographically and economically diverse African countries from the Economic Transformation Database: Botswana, Cameroon, Egypt, Ghana, Kenya, Malawi, Morocco, Mozambique, Namibia, Rwanda, Senegal, South Africa, and Zambia.[3] All countries in the sample exhibit peak manufacturing employment shares below approximately 15 percent, placing them within the premature deindustrialization regime identified by Rodrik (2016).

Our primary data source is the UNU-WIDER Economic Transformation Database (ETD),

[3] The ETD contains a slightly broader set of 21 African countries. Because our objective is to evaluate the mechanisms implied by the model, the discussion focuses on a subset of 13 countries for which the four-sector calibration most closely reproduces observed labour allocation patterns. This allows for a more concise presentation of the results while maintaining a geographically and economically diverse sample.

which is widely used in the structural transformation literature. This resource reports output (real and nominal gross value added),[4] and employment numbers for 51 countries across 12 sectors: agriculture, manufacturing, construction, mining, utilities, trade services, business services, transport services, financial services, government services, real estate, and other services. We self-construct sectoral labour productivities using output and employment.[5] As part of our extension from a three- to a four-sector framework, we decompose services into consumer and other services. We proxy the newly identified consumer services sector with the database's trade services sector, which is often grouped within the broader "distribution services" sector in alternative classifications (UNCTAD, 2004). However, because retail trade primarily serves final demand, researchers frequently use retail and distribution activities as a proxy for consumer-facing services (Peters et al., 2026). We have decided to include all other non-agriculture and non-manufacturing sub-sectors in our other services sector composite, including construction, mining and utility. Although their sectoral classification is debated, evidence from Africa suggests these sectors are highly service-intensive, relying heavily on engineering, logistics, maintenance, technical support, and other business services (Leeuw and Mtegha, 2016; Ramdoo and Cosbey, 2019).

We complement the ETD with a small number of additional international datasets used to construct calibration parameters and benchmark macroeconomic variables. In particular, we draw on the Penn World Tables for measures of real GDP, investment, physical and human capital; and deflators, exchange rates, and relative price levels, which are used to construct PPP conversion factors. We also use food expenditure share data from the World Bank's International Comparison Program to calibrate the agricultural subsistence parameter used in our model.

4 Stylized Facts

Three facts motivate our four-sector framework. Together they establish that Africa's structural transformation is neither the canonical industrialization-led path nor a uniform services expansion, but a heterogeneous process whose drivers vary substantially across countries and cannot be identified without disaggregating the service sector.

[4] Value added is the most common output measure used for measuring productivity due to lower data requirements (Eldridge and Powers, 2023).

[5] The Hodrick–Prescott filter is applied and used to extract the long-run trend component of selected series: real value added, employment numbers and labour productivity. This thereby smooths short-run volatility, and as we are analyzing annual data, we apply a smoothing parameter of $\lambda = 100$.

Labour exits agriculture without passing through manufacturing. Across our sample, the agricultural employment share falls on average from approximately 67 percent in 1990 to 49 percent by 2018 (Figure 2). Notably absent is the canonical Kuznets pattern in which labour flows first into manufacturing before shifting toward services (as identified in Figure 1). Manufacturing employment shares remain low and flat throughout, averaging approximately 7 percent by end of sample, with no evidence of the upward phase that historically preceded service expansion in today's advanced economies. This is consistent with the premature deindustrialization literature: manufacturing is no longer functioning as the broad-based engine of labour absorption it was for earlier industrializers.

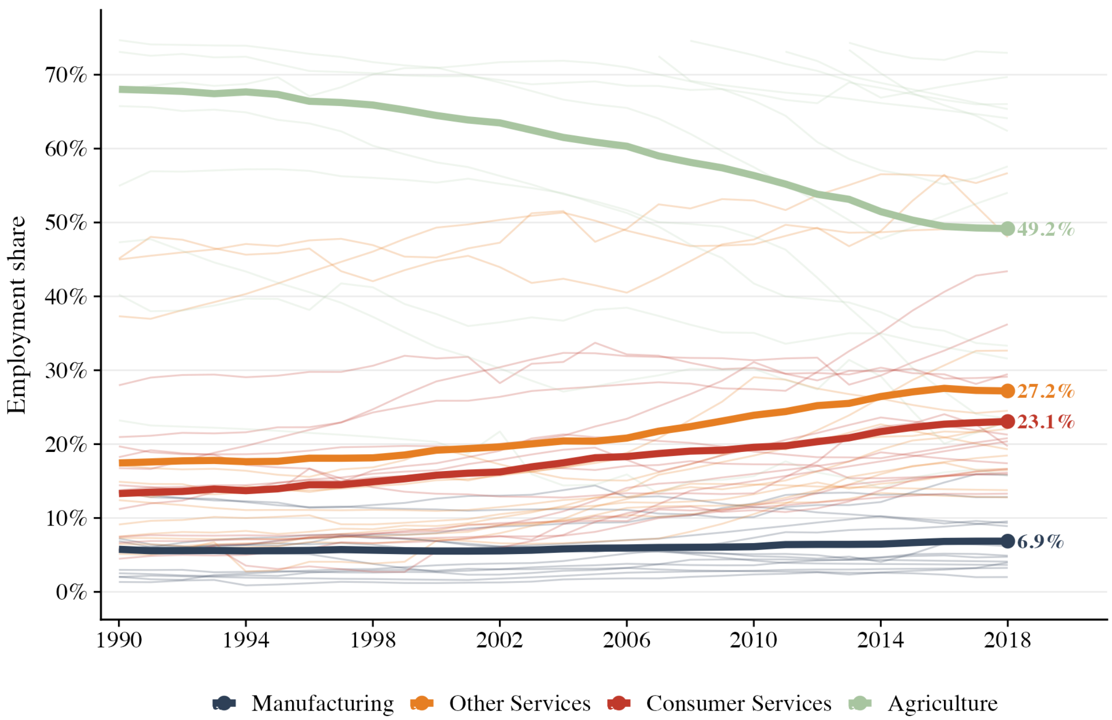

Figure 2: Average sectoral labour shares across our sample, 1990-2018
Source: UNU-WIDER ETD

Consumer services are the dominant absorber of reallocated labour: The sector absorbing labour released from agriculture is not manufacturing but consumer services (Figure 2). The consumer services employment share rises from roughly 14 percent in 1990 to 23 percent by 2018 — the largest absolute gain of any non-agricultural subsector over the period. Other services also expands, from around 16 to 27 percent, though this aggregate represents the cumulative growth of nine sub-sectors. The pattern holds broadly across countries, with meaningful variation in both levels and rates of change. Frameworks that treat services as a single residual sector therefore risk obscuring the variation most relevant to explaining structural change.

Consumer service employment growth masks substantial heterogeneity in productivity: Despite the near-universal rise in consumer services employment shares, underlying productivity trajectories are diverse across our fourteen-country sample (Figure 3). Normalizing value added per worker in consumer services to one in the initial year, India — the archetypal case of service-led growth (Fan et al., 2023) — records a roughly fourfold productivity increase between 1990 and 2018. While more than half of African countries also show gains, others see productivity stagnate or decline even as their employment shares rise. This divergence is the central empirical puzzle motivating our analysis: the same employment trend appears to be driven by productivity-pull in some countries and by labour absorption without productivity gains in others, a distinction the standard three-sector model cannot identify. Our four-sector extension is designed precisely to separate these two channels.

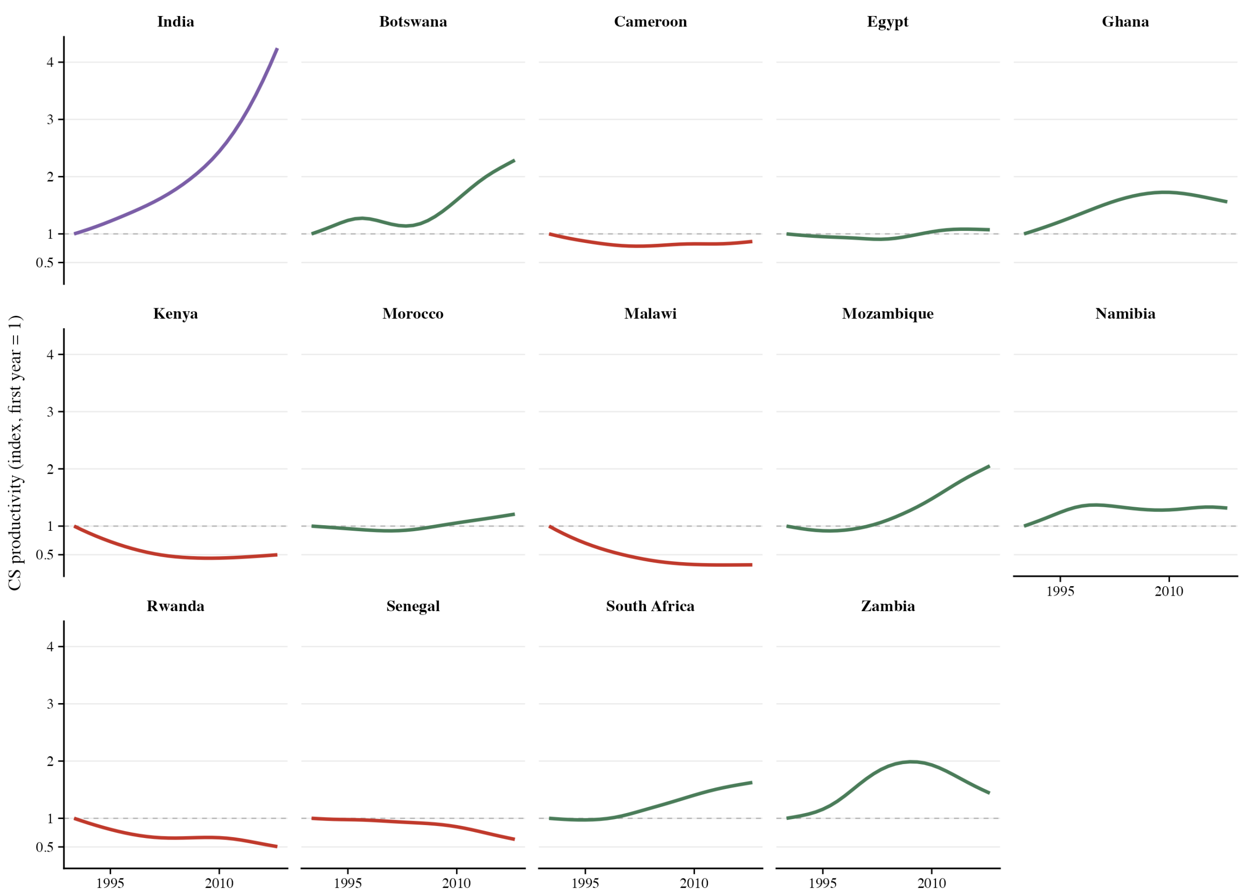

Figure 3: Consumer Service Productivity Heterogeneity
Source: Author's calcuations on UNU-WIDER ETD data

5 Methodology: Theoretical Framework & Calibration

5.1 Theoretical Framework

Our theoretical framework extends the three-sector model of Duarte and Restuccia (2010) by separating services into consumer services and other services. This modification allows us to examine whether different types of service activity contribute differently to labour reallocation and structural transformation through income and substitution effects.

Before introducing our extension, we apply the original Duarte and Restuccia (2010) framework in the African context. The benchmark model performs poorly when applied directly, though an Africa-specific recalibration substantially improves fit and generates feasible equilibria for all countries (Appendix Table B.1). Nevertheless, because services remain a single residual sector, the model cannot capture heterogeneity within the service economy or identify the distinct roles of consumer and other services. These limitations motivate the four-sector framework developed below; calibration details are discussed in the next section.

5.1.1 Production and the Firm's Problem

The economy consists of four sectors: agriculture (a), manufacturing (m), consumer services (cs), and other services (os), and a sector's production function is given by:

$$Y_{i,t} = A_{i,t} L_{i,t}, \qquad i \in \{a,\, m,\, cs,\, os\} \tag{1}$$

where $Y_{i,t}$ is output in sector i, $L_{i,t}$ is sectoral labour and $A_{i,t}$ is sector-specific productivity.[6] We assume the representative firm behaves competitively and sets marginal product equal to the wage w from their maximization problem. With the wage normalized to one, sectoral prices, $p_{i,t}$ are pinned down by productivity:

$$p_{i,t} = \frac{1}{A_{i,t}} \tag{2}$$

5.1.2 Preferences and the Household's Problem

We assume an infinitely-lived household with an endowment of L labour units (population normalized to 1) and log preferences over agricultural consumption and a non-agricultural

[6] Given that labour is the only factor of production in this model, labour productivity is analogous to total factor productivity (TFP).

composite good. The household solves lifetime utility:

$$\sum_{t=0}^{\infty} \beta^t \, u(c_{a,t},\, c_{N,t}), \qquad u(c_{a,t},\, c_{N,t}) = \alpha_a \log(c_{a,t} - \bar{a}) + (1-\alpha_a) \log(c_{N,t}) \tag{3}$$

where $\beta, \alpha_a \in (0,1)$ capture patience and agricultural preference share respectively. Agriculture includes an externally fixed agricultural subsistence level, $\bar{a}$, to account for the push effect – as the household becomes richer, agricultural expenditure share falls. The non-agricultural composite good $c_{N,t}$ is a constant elasticity of substitution (CES) aggregate of manufacturing, consumer services and other services, combining all three non-agricultural sectors within a single nest, extending Duarte and Restuccia (2010) as they explore two sectors within this nest. Like agriculture, consumer and other services each carry a Stone-Geary shifter (minimum subsistence requirement) to allow for differential income effects within the service sub-sectors.[7] The composite is defined as:

$$c_{N,t} = \Big[\alpha_m \, c_{m,t}^{\rho} + \alpha_{cs}(c_{cs,t} + \bar{cs})^{\rho} + \alpha_{os}(c_{os,t} + \bar{os})^{\rho}\Big]^{1/\rho}, \qquad \rho < 1 \tag{4}$$

where ρ governs substitution between manufacturing, consumer services and other services across the single non-agricultural nest. Preference shares for manufacturing (α_m), consumer services (α_{cs}) and other services (α_{os}) lie between 0 and 1 with $\alpha_m + \alpha_{cs} + \alpha_{os} = 1$. The shifters $\bar{cs} \geq 0$ and $\bar{os} \geq 0$ generate income elasticities that exceed one for consumer and other services respectively. When income increases, more resources are allocated to these service sectors. The representative household maximizes utility subject to the budget constraint:[8]

$$p_{a,t}c_{a,t} + p_{m,t}c_{m,t} + p_{cs,t}c_{cs,t} + p_{os,t}c_{os,t} = wL_t \tag{5}$$

Goods and labour market clearing conditions require:

$$c_{i,t} = Y_{i,t} = A_{i,t}L_{i,t} \tag{6}$$

$$L_{a,t} + L_{m,t} + L_{cs,t} + L_{os,t} = L_t \tag{7}$$

From competitive equilibria, first-order conditions from the household's maximization problem and market clearing conditions, we generate equilibrium demand and labour allocation

[7] Stone-Geary preferences depart from standard homothetic utility by introducing a subsistence floor on agriculture ($\bar{a}$) and luxury shifters on consumer services ($\bar{cs}$) and other services ($\bar{os}$), implying non-linear Engel curves that generate labour reallocation from agriculture to services as income grows, even absent productivity differences.

[8] In line with Duarte and Restuccia (2010), the absence of capital in the model makes this sequence of static problems, meaning we are able to abstract from intertemporal decisions.

conditions, first for agriculture:

Agriculture: Our equilibrium agricultural labour allocation is given by:

$$L_{a,t} = (1-\alpha_a)\,\frac{\bar{a}}{A_{a,t}} + \alpha_a\left(L_t + \frac{\bar{cs}}{A_{cs,t}} + \frac{\bar{os}}{A_{os,t}}\right) \tag{8}$$

The first term captures the imposition of agricultural subsistence on labour share: as agricultural productivity rises this cost falls and workers leave agriculture. The second term captures the agricultural preference share α_a applied to full surplus income (minimum services thresholds). The income effect further reduces the agricultural share as rising aggregate income drives demand more toward non-agricultural goods. As sectoral productivities increase, however, agricultural labour share asymptotically converges to its preference share, α_a.

Manufacturing, consumer services, and other services: For the non-agricultural sectors, their relative demands are governed by relative productivities and our elasticity parameter. First-order conditions yield pairwise relative demand conditions for any two non-agricultural sectors. For example, consumer services relative to manufacturing yields:

$$\frac{\alpha_{cs}(c_{cs,t}+\bar{cs})^{\rho-1}}{\alpha_m\, c_{m,t}^{\rho-1}} = \frac{p_{cs,t}}{p_{m,t}} \tag{9}$$

By substituting in our competitive and market clearing equilibria, the relative demand conditions translate into relative labour allocation conditions. For relative consumer services–manufacturing:

$$\frac{A_{cs,t}L_{cs,t}+\bar{cs}}{A_{m,t}L_{m,t}} = \left(\frac{\alpha_m}{\alpha_{cs}}\cdot\frac{A_{m,t}}{A_{cs,t}}\right)^{\frac{1}{\rho-1}} \tag{10}$$

Labour allocation between non-agricultural sectors is inversely related to relative prices and positively related to relative productivities, with the strength of reallocation governed by ρ. When $\rho < 0$, sectors are gross complements — even as manufacturing goods become cheaper through rising productivity, demand for services increases, consistent with the observed patterns of structural transformation in the literature. The Stone–Geary shifters $\bar{cs}$ and $\bar{os}$ generate an additional income effect within the non-agricultural nest: as aggregate income rises, a disproportionate share of expenditure flows toward consumer and other services and away from manufacturing, even holding relative productivities fixed. This is what we would expect in countries with lower manufacturing labour shares.

5.2 Calibration

We assess our model's ability to explain structural change by estimating paths for labour shares given selected parameter values and compare this to real-life data values. We do this for a selected sample of 13 African countries, comparing the results to the case of India. Observed sectoral productivity paths are fed into the model and then for each country-year, the model computes predicted labour allocations from our equations above:

$$\hat{L}_{i,t}, \qquad i \in \{a, m, cs, os\} \tag{11}$$

$L_{a,t}$ is obtained directly from equation (8) in closed form, after which $\{L_{m,t}, L_{cs,t}, L_{os,t}\}$ are jointly determined by the sectoral wage-equalisation conditions and labour market clearing (7). When $\bar{cs} = \bar{os} = 0$ this second step is fully closed form; and with the Stone–Geary shifters, a single non-linear equation in $L_{m,t}$ remains, which is solved numerically.

Within each country, we normalize productivity levels across sectors to manufacturing productivity in 1990, setting manufacturing productivity equal to one and expressing all other sectoral productivity levels relative to this benchmark.[9] With productivity growth rates ($\gamma_{i,t}$ real sectoral value added per worker), we obtain a path for sectoral productivity such that: $A_{i,t+1} = (1 + \gamma_{i,t})A_{i,t}$. Parameters are reported in Table 1 and have been informed by literature or estimated/proxied with real-life data:

- **Elasticity of substitution** (ρ) is set at -0.5. Its negative value is consistent with Duarte and Restuccia (2010)'s calibration ($\rho = -1.5$), while its smaller magnitude reflects calibration of both the three- and four-sector models to our country sample.

- **Preference shares** (α_i) are informed by the cross-country average of sectoral labour shares in 2018. In line with Duarte and Restuccia (2010), these are treated as medium- to long-run preferences. For instance, the agriculture preference share (α_a) is 0.30 although the 2018 sample average is 42 percent.

- **Stone-Geary shifters** for other services and consumer services are computed according to respective non-agricultural labour shares in 1990.

- **Agricultural subsistence** ($\bar{a}$) is computed using country-specific food expenditure shares (calculation is reported in Appendix A.2).

[9] We choose manufacturing as the reference sector because it exhibits the least cyclical volatility and is least susceptible to measurement error.

Table 1: List of Parameters for Calibration

Parameter	Value	Source
$\bar{a}$	0.02–0.26	Food expenditure shares (country-specific)
$\bar{cs}$	0.10	Approximate labour share of consumer services in 1990
$\bar{os}$	0.20	Approximate labour share of other Services in 1990
ρ	-0.5	African-specific three-sector model calibration
α_a	0.30	Long-run sectoral labour share
α_m	0.04	Long-run sectoral labour share
α_{cs}	0.35	Long-run sectoral labour share
α_{os}	0.61	Residual

Note: See Appendix Tables 11 and 12 for robustness checks on ρ and the subsistence parameters.

We then run counterfactual exercises to explore different pathways if certain variables are fixed or set alternatively, as discussed in Section 6.2

6 Results

6.1 Model Calibration

Using a common parameterization across countries, the four-sector model successfully generates feasible equilibria and predicts labour allocation patterns for the entire sample (root mean squared errors reported in Table 2). Relative to the Africa-calibrated three-sector benchmark, average RMSE falls from 0.118 to 0.109, improving fit by approximately 7.5 while keeping model parameters fixed. The resulting specification captures greater service-sector heterogeneity while remaining parsimonious. Nevertheless, model performance continues to vary across countries. For example, the model underpredicts agricultural labour shares by an average of 12.5 percentage points in Ghana and 25.4 percentage points in Rwanda (Figure 4).

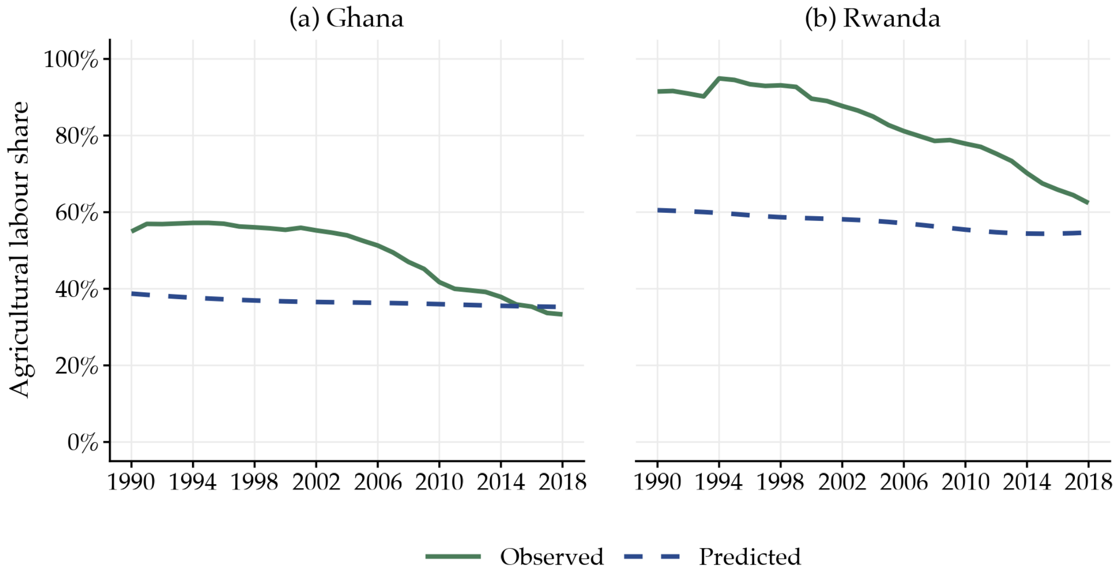

Figure 4: Agricultural Labour Share: Predicted (model) vs Observed (data)
Source: UNU-WIDER ETD and authors' calculations.

The key result is that disaggregating services into consumer and other services yields a meaningful improvement in explanatory power relative to the aggregate-services benchmark. This suggests that heterogeneity within the service sector is an important component of structural transformation and that treating services as a single residual category obscures economically relevant differences in labour allocation dynamics.

Table 2: Four-Sector Results: Root Mean Squared Errors (RMSE)

Country	RMSE	Country	RMSE
Morocco	0.053	Senegal	0.106
Mozambique	0.072	Cameroon	0.123
Kenya	0.084	Namibia	0.124
Malawi	0.089	Zambia	0.127
Ghana	0.102	India	0.132
Botswana	0.103	South Africa	0.155
Egypt	0.103	Rwanda	0.156

Notes: The calibration uses $\alpha_a = 0.30$, $\alpha_m = 0.04$, $\alpha_{cs} = 0.35$, $\rho = -0.5$, and country-specific $\bar{a}$. All countries generate feasible equilibria over the full sample period. RMSE is calculated using predicted and observed sectoral labour shares across all available years.

6.2 Counterfactual Analysis

We construct five counterfactuals (CFs) to isolate the quantitative importance of sectoral productivity and luxury demand (collectively referred to as channels) in driving structural transformation, which we frame as the following questions:

- **Fixing sectoral productivity:** What would the path of labour allocation be if different sectoral productivities were fixed at their 1990 level, shutting down the relative price and income effects that arise from productivity changes in that sector? We hold consumer services (CF1), other services (CF3) and agricultural (CF4) productivity flat respectively: $A_{i,t} = A_{i,1990}$.
- **Removing income effects:** What would the path of labour allocation be in the absence of income effects, i.e. $\bar{cs} = 0$? (CF2)
- **Using India's consumer services productivity growth:**[10] What would the path of labour allocation be if the African countries in our sample had experienced India's consumer services productivity growth? (CF5)

Comparing the baseline to each counterfactual isolates the contribution of each channel to structural transformation (see Appendix Table 6 for their construction). By differencing CF2 and CF1, we further decompose the effect of consumer services productivity growth into income and substitution effects. These decompositions allow us to identify and compare the leading drivers of structural transformation across our sample.[11] Table 3 reports counterfactual results for Ghana, Rwanda, and India; full results appear in Appendix Table 7. In all three, the income effect pulls labour toward consumer services, raising consumer service labour share by 3.4 percentage points in Ghana and 6.9 percentage points in India. Rwanda moves in the opposite direction: consumer service productivity growth is negative, reducing consumer service labour share by 5.2 percentage points, and while the income effect is larger (9.8 percentage points), it cannot offset this drag. The contrast therefore reflects not the size of income effects, which are large in all three cases, but the direction of consumer service productivity growth: where positive, income effects amplify structural transformation; where negative, they cannot reverse the overall drag.

[10] CF5 rescales India's consumer services productivity growth path to each country's own 1990 baseline, capturing differences in growth rates rather than levels, which are not comparable across countries due to independent country-level normalization.

[11] For completeness, Appendix Table 9 documents the effect of fixing A_m at its 1990 level on within-non-agricultural reallocation; the magnitudes are small (below 2 percent for most countries), confirming that manufacturing productivity plays a limited role across the sample.

Table 3: Counterfactual Contributions to Labour Reallocation: Ghana, Rwanda, and India

	Agriculture	Manufacturing	Consumer Services (CS)	Other Services (OS)
Panel A: Ghana				
CF1: CS productivity growth	−1.5	0.2	0.3	1.0
CF2: CS income effect	−1.5	−0.3	3.4	−1.5
Substitution effect (CF2−CF1)	0.0	0.5	−3.0	2.5
CF3: OS productivity growth	−0.7	0.2	1.1	−0.6
CF4: Agricultural push	−1.3	0.1	0.6	0.6
CF5: India's CS growth path	−1.6	0.7	−2.2	3.1
Panel B: Rwanda				
CF1: CS productivity growth	4.4	0.1	−5.2	0.7
CF2: CS income effect	4.4	0.9	−9.8	4.5
Substitution effect (CF2−CF1)	0.0	−0.8	4.6	−3.8
CF3: OS productivity growth	0.4	0.0	0.1	−0.4
CF4: Agricultural push	−10.6	0.9	4.6	5.2
CF5: India's CS growth path	−7.9	0.4	4.9	2.6
Panel C: India				
CF1: CS productivity growth	−3.0	0.7	−2.4	4.7
CF2: CS income effect	−3.0	−0.7	6.9	−3.2
Substitution effect (CF2−CF1)	0.0	1.4	−9.3	7.9
CF3: OS productivity growth	−2.1	0.6	2.9	−1.4
CF4: Agricultural push	−1.6	0.1	0.6	0.8
CF5: India's CS growth path	0.0	0.0	0.0	0.0

Note: These figures report percentage point changes. A positive value indicates the channel increased that sector's labour share; a negative value indicates it reduced it.

Figure 5 presents a varied picture of the scope and drivers of agricultural exit across our sample, again with Ghana, Rwanda, and India illustrating three distinct cases.[12] Ghana and India are consumer-services-pull economies: consumer services productivity growth is the dominant driver of agricultural exit, responsible for 43 and 45 percent of observed exit respectively. In India, other services productivity growth contributes a further 32 percent, consistent with its service-led characterization. In Ghana, agricultural productivity improvements play the second-largest role, contributing 36 percent. Manufacturing plays a negligible role here and across our whole sample.

[12] The signs and relative magnitudes of the income and substitution effects underlying this decomposition are stable across alternative parameterizations of ρ, $\bar{cs}$, and $\bar{os}$ (see Appendix Tables 11 and 12).

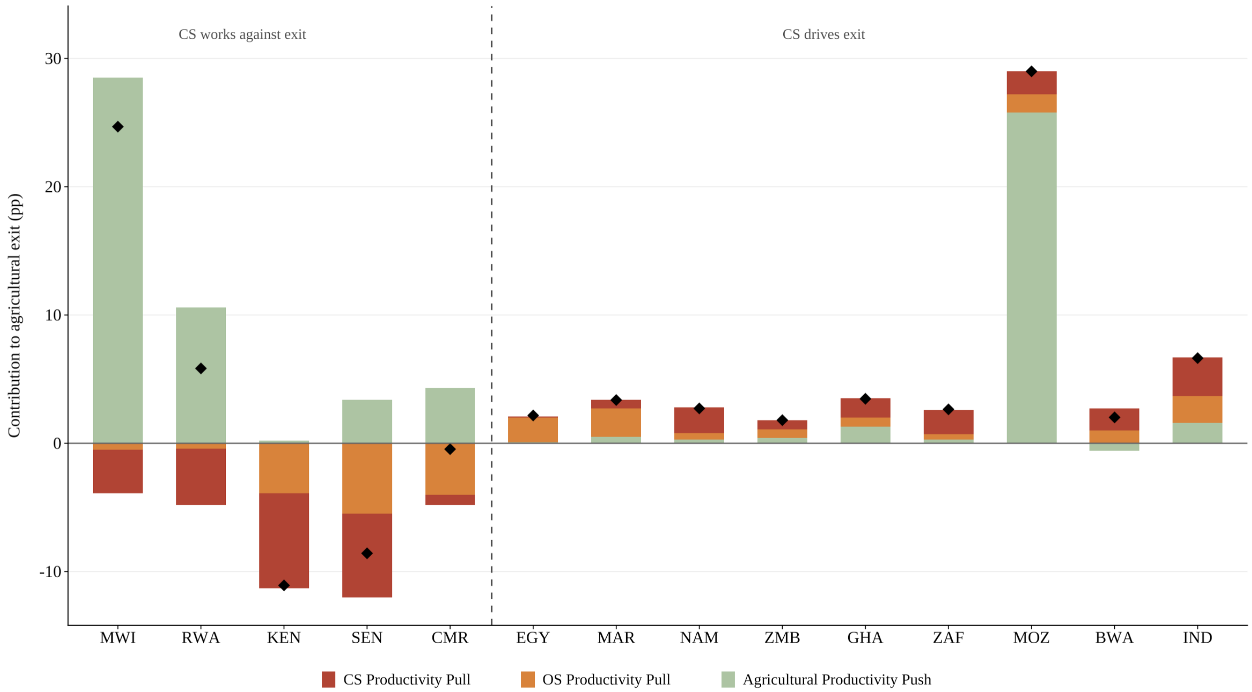

Figure 5: Decomposition of Agricultural Labour Exit

Source: Authors' calculations. Note: Negative CS productivity growth countries fall on the left of the dotted line, and positive CS productivity growth countries on the right. Black diamonds indicate observed agricultural exit.

These exercises allow us to identify four broad groups. In a first group (Mozambique, Malawi, and Rwanda), agricultural push dominates, accounting for over 89% of observed exit respectively (country-level percentages reported in Appendix Table 8). Rwanda is further distinguished by consumer services productivity growth impeding structural transformation: agricultural labour share would have been 4.4 percentage points lower had consumer services productivity remained at its 1990 level (Table 3). In a second group (Egypt, Morocco, and Zambia), labour leaving agriculture is absorbed primarily through other services productivity growth rather than consumer services. In a third group (Kenya, Senegal, and, to a lesser extent, Cameroon), negative consumer services productivity growth is slowing the overall pace of structural transformation with these countries having a higher agricultural labour share by the data horizon. In a final group (Ghana, Botswana, Namibia, and South Africa), sustained consumer services productivity growth is the largest single contributor to agricultural exit. That this emerges as the dominant driver for four African economies in our sample points to a dimension of structural transformation that remains under-examined in both academic and the policy spaces.

Figure 6 shows the additional percentage point increase in agricultural exit that would occur if our African sample had the consumer services productivity growth rate of India, and it demonstrates how strong consumer services productivity growth can accelerate structural transformation. Under India's consumer services productivity growth path (of more than four fold over the period, shown by Figure 1), every country in the sample would have experienced faster agricultural exit. While it follows mechanically from the counterfactual's construction that countries with the weakest consumer services productivity see the largest gains, this nonetheless demonstrates that the variation in structural transformation outcomes across our sample reflects differences in consumer services productivity growth rates rather than immutable structural characteristics. This has important implications for policy and broader debates on economic transformation, which we turn to next.

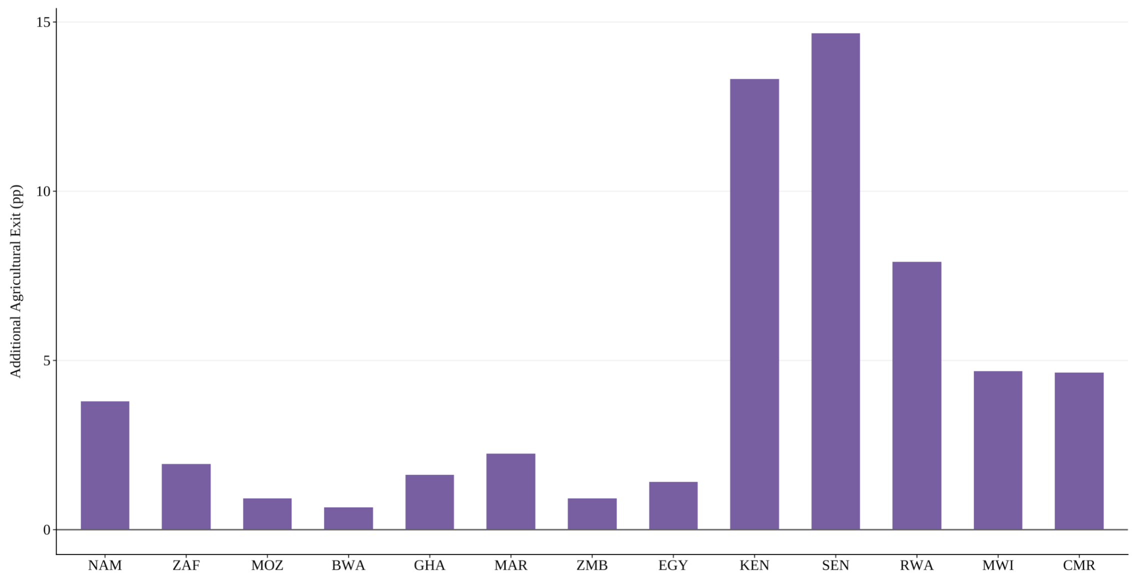

Figure 6: Additional Agricultural Labour Exit with India's CS Productivity Growth
Source: Authors' calculations.

7 Discussion

Our results suggest that Africa is not experiencing a single pattern of structural transformation following premature deindustrialization. While consumer services productivity is an important driver of agricultural exit in some countries, in others where consumer services productivity is weak or nonexistent, structural transformation remains dominated by agricultural productivity growth or expansion of other services (Figure 7). In some others, we observe minimal movement out of agriculture (consumer services-drag). This heterogeneity has important implications for how we interpret premature deindustrialization and the role of services in development and these patterns suggest that the decline of manufacturing has not produced a single alternative development trajectory, but rather a diverse set of pathways through the service economy. This has implications for policy and further extensions.

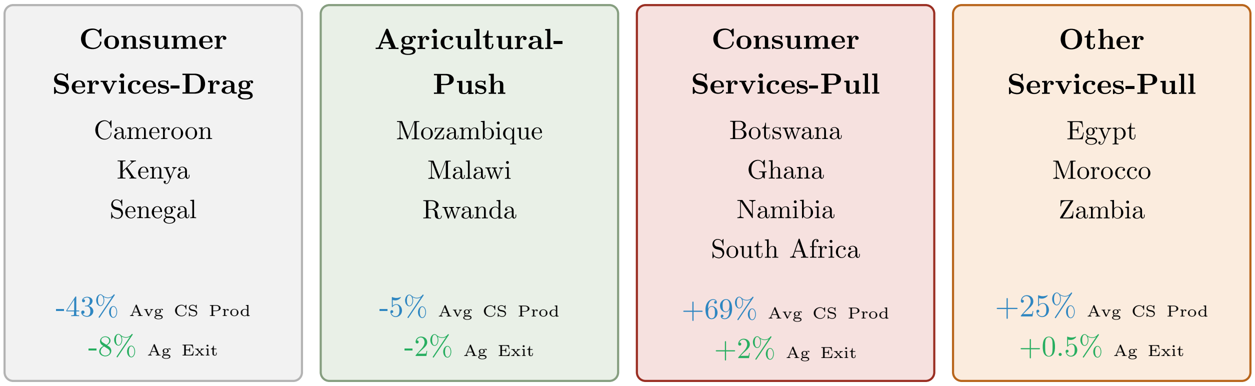

Figure 7: Four Pathways of Structural Transformation in the Sample

Note: Countries are grouped by dominant mechanism. Avg CS Prod denotes average consumer services productivity growth; Ag Exit denotes the average contribution of consumer services productivity to agricultural employment decline.

7.1 Policy Implications

A first implication of our results is that policymakers should distinguish between service-sector expansion and productivity growth within services. As a first step beyond the model, we run a reduced-form illustrative regression of GDP growth on consumer service productivity growth and employment shares.[13] Across multiple fixed-effects specifications, consumer service productivity growth is positively associated with GDP growth, while employment shares are not — suggesting that developmental significance depends less on sector size than on productivity performance. Although these relationships are descriptive rather than causal, the expansion of consumer services employment should not automatically be interpreted as evidence of productivity-enhancing structural transformation.

[13]See details in Appendix 10.

Table 4: GDP Growth and Consumer Services Productivity/Employment

	Country & Year Fixed Effects	Urban Controls	Country Fixed Effects
CS productivity growth	0.095***	0.088***	0.092***
	(0.023)	(0.027)	(0.029)
CS employment share	0.225	0.323*	0.322
	(0.154)	(0.155)	(0.187)
Observations	364	334	334
Adj. R^2	0.231	0.293	0.229

Notes: For regression specifciations, the dependent variable is annual GDP growth with the main regressor being either CS productivity growth or employment shares. Growth rates are measured as log differences. Standard errors clustered at the country level. All specifications include investment and human capital controls; Column (2) additionally includes urbanization controls. Source: Authors' calculations based on PWT and ETD. Full results reported in Appendix Table 10.

This distinction has important policy implications. The objective is not a larger consumer services sector per se, but a more productive one — productivity gains matter insofar as they generate the income growth needed to sustain demand expansion. Policies improving competition, technology adoption, infrastructure quality, and access to finance within consumer services are therefore more consequential than those aimed simply at facilitating labour reallocation into services. Evidence from Ghana and South Africa suggests that growing digitalization, including mobile-money adoption and retail self-service technologies, has already enhanced efficiency in consumer services (Turkson et al., 2025; Tshivhase, 2023), with potential spillover effects for the broader economy.

Our finding that declining consumer services productivity drags back structural transformation has additional implications for policymakers. Measures to improve agricultural productivity have been a dominant focus of both public policy and private sector organizations across Africa, and our results suggest this focus is well-founded: agricultural push is the dominant driver of exit in several countries in our sample. However, its effectiveness is conditional on consumer services productivity. Where consumer services productivity is weak or declining, the income and price effects that would otherwise amplify the benefits of agricultural productivity growth are dampened, offering a potential explanation for why agricultural productivity interventions have not always yielded anticipated results (Wollburg, 2024; Owoo, 2026). Understanding consumer service productivity may therefore be as important as understanding agriculture itself for predicting whether structural transformation takes hold. Raising consumer services productivity is not an alternative to agricultural policy, but a necessary complement and, though preliminary, our regression results suggest it is consumer services productivity growth, rather than the size of the sector, that is associated with stronger economic performance.

Additionally, as consumer services are capable of absorbing large numbers of workers, policies that improve productivity in activities such as retail, transport, and other consumer-facing services may increase migration toward urban areas. Evidence from Africa suggests that productivity-enhancing interventions can generate substantial employment growth in service activities (Hjort and Poulsen, 2019), while urbanization and agglomeration may further reinforce these productivity gains (Peters et al., 2026). Complementary investments in urban infrastructure, public services, and labour-market integration may therefore be required to ensure that productivity gains translate into improved living standards rather than urban congestion or labour-market exclusion. More broadly, the benefits of productivity growth in consumer services may depend not only on the productivity gains themselves but also on the capacity of urban institutions to absorb and support a growing workforce.

7.2 Limitations and Extensions

Several limitations suggest avenues for future research. First, the model assumes a closed economy, following the structural transformation literature (Kongsamut et al., 2001; Ngai and Pissarides, 2007; Duarte and Restuccia, 2010). This is more defensible for our sample than for those typically studied in this literature, given that Africa accounts for just 3 per cent of global trade in goods and services (Coulibaly et al., 2022). Nonetheless, export opportunities, import competition, and global value chain integration can shape productivity growth and labour allocation (de Vries et al., 2026), and extending the framework to an open-economy setting remains a natural avenue for future work.

Second, our other services category remains a broad residual encompassing construction, finance, and business services. Disaggregating this category could reveal important mechanisms within the service economy, and business services in particular may be an important source of productivity-enhancing structural transformation in developing economies (Sen, 2023).

Third, the model assumes a single elasticity of substitution across all non-agricultural sectors. Substitution patterns likely differ across manufacturing, consumer services, and other services, and a nested demand structure allowing for greater flexibility would be a natural extension.

Fourth, our calibration spans a relatively short time horizon (1990–2018), which may limit our ability to identify stable long-run preference parameters. Longer time series, as they become available for African economies, would allow for more robust identification of preference parameters and a richer characterization of long-run structural transformation dynamics.

Finally, the ETD records formal sectoral employment, yet non-agricultural informal employment accounts for over 56 per cent of total employment in Africa (ILO, 2018). While informal work is conventionally associated with low productivity, recent evidence suggests that productivity gains within the informal sector may themselves be driving consumer service productivity growth in Africa (Peters et al., 2026). By how much informality drags or adds to consumer service productivity, and how sensitive our findings are to the formal-sector productivity measures the ETD provides, remains an important avenue for future research.

8 Conclusions

By extending the Duarte-Restuccia framework to distinguish consumer from other services and applying it to thirteen African countries and India, we find substantial heterogeneity in the drivers of structural transformation, with consumer services productivity acting as a major engine of agricultural exit in some economies but suppressing it in others.

Our results highlight the importance of distinguishing between service-sector expansion and productivity growth within services. While consumer services increasingly absorb labour across much of Africa, the underlying drivers of this process differ substantially across countries. In some cases, consumer services contribute to sustained productivity growth and labour reallocation; in others, they function primarily as a repository for workers displaced from agriculture.

More broadly, these findings suggest that "skipping the factory" need not imply a common development trajectory. The decline of manufacturing as a major source of employment has not produced a uniform shift into consumer services, nor are all transitions into service-sector employment equally productivity enhancing. Understanding the conditions under which informal consumer services enhance or hinder aggregate productivity would both advance the literature and offer clearer guidance for policymakers seeking to harness consumer service growth in driving development outcomes.

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9 Appendix

A Methodological Expansions

A.1 Full Theoretical Model Derivation

This appendix derives the equilibrium labour allocation conditions underlying the calibration. Production, preferences, and market clearing are as set out in Section 5; we collect the key objects here and work through the derivation in full.

Lagrangian and First-Order Conditions

The household's static problem is to maximise utility subject to the budget constraint. With λ as the multiplier, the Lagrangian is:

$$\mathcal{L} = \alpha_a \log(c_{a,t} - \bar{a}) + (1-\alpha_a)\log(c_{N,t}) + \lambda\big(wL_t - p_{a,t}c_{a,t} - p_{m,t}c_{m,t} - p_{cs,t}c_{cs,t} - p_{os,t}c_{os,t}\big) \quad (12)$$

Differentiating with respect to each consumption good yields the following first-order conditions (FOCs):

Agriculture: $\dfrac{\alpha_a}{c_{a,t} - \bar{a}} = \lambda p_{a,t}$

Manufacturing: $(1-\alpha_a)\, c_{N,t}^{-\rho}\, \alpha_m\, c_{m,t}^{\rho-1} = \lambda p_{m,t}$

Consumer services: $(1-\alpha_a)\, c_{N,t}^{-\rho}\, \alpha_{cs}(c_{cs,t} + \bar{cs})^{\rho-1} = \lambda p_{cs,t}$

Other services: $(1-\alpha_a)\, c_{N,t}^{-\rho}\, \alpha_{os}(c_{os,t} + \bar{os})^{\rho-1} = \lambda p_{os,t}$

Relative Demand Conditions

Dividing the consumer services FOC by the manufacturing FOC cancels $(1-\alpha_a)$, $c_{N,t}^{-\rho}$, and λ and derives:

$$\frac{\alpha_{cs}(c_{cs,t} + \bar{cs})^{\rho-1}}{\alpha_m\, c_{m,t}^{\rho-1}} = \frac{p_{cs,t}}{p_{m,t}} \quad (13)$$

with an analogous expression for other services. Dividing the agricultural FOC by the sum of non-agricultural FOCs and rearranging gives the agriculture–non-agriculture expenditure ratio:

$$\frac{p_{a,t}(c_{a,t}-\bar{a})}{p_{m,t}c_{m,t}+p_{cs,t}(c_{cs,t}+\bar{cs})+p_{os,t}(c_{os,t}+\bar{os})}=\frac{\alpha_a}{1-\alpha_a} \tag{14}$$

Equilibrium Labour Allocation

Substituting the competitive pricing condition $p_{i,t} = 1/A_{i,t}$ and goods-market clearing $c_{i,t} = A_{i,t}L_{i,t}$ into equation 14 yields the closed-form agricultural labour allocation:

$$L_{a,t} = (1-\alpha_a)\frac{\bar{a}}{A_{a,t}} + \alpha_a\left(L_t + \frac{\bar{cs}}{A_{cs,t}} + \frac{\bar{os}}{A_{os,t}}\right) \tag{15}$$

Applying the same substitutions to the non-agricultural relative demand conditions and raising both sides to the power $\frac{1}{\rho-1}$ gives:

$$\frac{A_{cs,t}L_{cs,t}+\bar{cs}}{A_{m,t}L_{m,t}} = \left(\frac{\alpha_m}{\alpha_{cs}}\cdot\frac{A_{m,t}}{A_{cs,t}}\right)^{\frac{1}{\rho-1}} \tag{16}$$

$$\frac{A_{os,t}L_{os,t}+\bar{os}}{A_{m,t}L_{m,t}} = \left(\frac{\alpha_m}{\alpha_{os}}\cdot\frac{A_{m,t}}{A_{os,t}}\right)^{\frac{1}{\rho-1}} \tag{17}$$

Together with labour market clearing $L_{a,t} + L_{m,t} + L_{cs,t} + L_{os,t} = L_t$, these three conditions jointly determine $\{L_{m,t}, L_{cs,t}, L_{os,t}\}$ given $L_{a,t}$. Since $\rho < 0$, the exponent $1/(\rho-1)$ is negative, so a rise in relative productivity A_{cs}/A_m raises the consumer-services labour share — sectors are gross complements.

A.2 Calculation of $\bar{a}$

To calibrate the agricultural subsistence parameter ($\bar{a}$), we use 2017 International Comparison Program (ICP) data from the World Bank. The calibration relies on two country-level variables: the food share of consumption and Actual Individual Consumption (AIC) per capita, both measured in PPP-adjusted 2017 US dollars.

Let f_j denote the food share of consumption in country j, and let AIC_j denote per-capita Actual Individual Consumption. We estimate:

$$f_j = \alpha + \beta\left(\frac{1}{AIC_j}\right) + \varepsilon_j$$

where β captures the implied subsistence food requirement in PPP-adjusted dollars. In our baseline estimation:

$$\hat{\beta} = 298.71$$

We then convert this common PPP-denominated subsistence requirement into a country-specific model parameter:

$$\bar{a}_j = \frac{\hat{\beta}}{AIC_j}$$

Poorer countries therefore receive a higher $\bar{a}_j$, as the same PPP-valued subsistence requirement represents a larger share of total consumption.

Country	$\bar{a}$	Country	$\bar{a}$
Botswana	0.037	Namibia	0.030
Cameroon	0.089	Rwanda	0.150
Egypt	0.022	Senegal	0.097
Ghana	0.062	South Africa	0.026
Kenya	0.075	Zambia	0.145
Malawi	0.196		
Morocco	0.049	India	0.103
Mozambique	0.256		

A.3 Counterfactual Construction

Table 6: Counterfactual Construction and Identification

	Description	Fixed element	Difference identifies
Baseline	Model's best fit to observed data	—	—
CF1	CS productivity at 1990 level	$A_{cs,t} = A_{cs,1990}$	Baseline − CF1: CS productivity contribution
CF2	CS income effect switched off	$\bar{cs} = 0$	Baseline − CF2: CS income effect
			CF2 − CF1: Substitution effect
CF3	OS productivity at 1990 level	$A_{os,t} = A_{os,1990}$	Baseline − CF3: OS productivity contribution
CF4	Agricultural productivity at 1990 level	$A_{a,t} = A_{a,1990}$	Baseline − CF4: Agricultural push
CF5	India's CS productivity growth path	$A_{cs,t} = A_{cs,1990} \cdot \frac{A_{cs,\text{India},t}}{A_{cs,\text{India},1990}}$	Baseline − CF5: Role of CS productivity growth rate

B Full Results

B.1 Three-Sector Model Results

Original DR Benchmark

Avg. RMSE = 0.194

Country	Feasible	RMSE
Botswana	No	–
Cameroon	No	–
Egypt	Yes	0.120
Ghana	Yes	0.206
Kenya	No	–
Morocco	Yes	0.122
Mozambique	Yes	0.247
Malawi	No	–
Namibia	No	–
Rwanda	No	–
Senegal	No	–
South Africa	No	–
Zambia	No	–
India	Yes	0.274

Africa-Calibrated Benchmark

Avg. RMSE = 0.118

Country	Feasible	RMSE
Morocco	Yes	0.035
Botswana	Yes	0.038
Egypt	Yes	0.077
Mozambique	Yes	0.084
Malawi	Yes	0.087
Kenya	Yes	0.110
Ghana	Yes	0.115
Namibia	Yes	0.134
Cameroon	Yes	0.141
Zambia	Yes	0.153
India	Yes	0.158
Senegal	Yes	0.165
Rwanda	Yes	0.171
South Africa	Yes	0.184

Notes: The left-hand panel reports results from the original Duarte and Restuccia (2010) benchmark calibration ($\alpha_a = 0.01$, $\alpha_m = 0.04$, $\rho = -1.5$, $\bar{s} = 0.89$, $\bar{a} = 0.11$). The right-hand panel reports an Africa-calibrated benchmark model using sample-based parameter values ($\alpha_a = 0.30$, $\alpha_m = 0.04$, $\rho = -0.5$, $\bar{s} = 0.40$, country-specific $\bar{a}$). "Feasible" indicates whether the model generates non-negative equilibrium labour allocations for all sectors over the sample period. RMSE is calculated over agriculture, manufacturing, and services labour shares for all years.

B.2 Four-Sector Model Results

Average RMSE = 0.109

Country	RMSE	Country	RMSE
Morocco	0.053	Senegal	0.106
Mozambique	0.072	Cameroon	0.123
Kenya	0.084	Namibia	0.124
Malawi	0.089	Zambia	0.127
Ghana	0.102	India	0.132
Botswana	0.103	South Africa	0.155
Egypt	0.103	Rwanda	0.156

Notes: Results are reported for the preferred four-sector specification comprising agriculture, manufacturing, consumer services, and other services. The calibration uses $\alpha_a = 0.30$, $\alpha_m = 0.04$, $\alpha_{cs} = 0.10$, $\alpha_{os} = 0.20$, $\rho = -0.5$, and country-specific $\bar{a}$. All countries generate feasible equilibria over the full sample period. RMSE is calculated using predicted and observed sectoral labour shares across all available years.

B.3 Counterfactual Results

Table 7: Contribution of Each Channel to Sectoral Labour Share Change, 1990–2018: All Countries

	Agric.	Manuf.	Cons. Services	Other Services
Panel A: Botswana				
CF1: CS productivity growth	−1.7	0.4	−1.1	2.4
CF2: CS income effect	−1.7	−0.4	3.9	−1.8
Substitution effect (CF2−CF1)	0.0	0.8	−5.0	4.2
CF3: OS productivity growth	−1.0	0.2	0.7	0.1
CF4: Agricultural push	0.6	−0.1	−0.2	−0.3
CF5: India's CS growth path	−0.7	0.4	−2.1	2.4
Panel B: Cameroon				
CF1: CS productivity growth	0.8	0.0	−0.7	0.0
CF2: CS income effect	0.8	0.1	−2.4	1.6
Substitution effect (CF2−CF1)	0.0	−0.1	1.7	−1.6
CF3: OS productivity growth	4.0	−0.2	−1.2	−2.6
CF4: Agricultural push	−4.3	0.4	1.8	2.2
CF5: India's CS growth path	−4.6	0.6	0.8	3.3

Continued on next page

Table 7 continued

	Agric.	Manuf.	Consumer Services	Other Services
Panel C: Egypt				
CF1: CS productivity growth	−0.1	0.0	−0.1	0.2
CF2: CS income effect	−0.1	−0.1	0.5	−0.3
Substitution effect (CF2−CF1)	0.0	0.1	−0.6	0.5
CF3: OS productivity growth	−1.9	0.2	0.8	1.0
CF4: Agricultural push	−0.1	0.0	0.0	0.1
CF5: India's CS growth path	−1.4	0.8	−4.5	5.1
Panel D: Ghana				
CF1: CS productivity growth	−1.5	0.2	0.3	1.0
CF2: CS income effect	−1.5	−0.3	3.4	−1.5
Substitution effect (CF2−CF1)	0.0	0.5	−3.0	2.5
CF3: OS productivity growth	−0.7	0.2	1.1	−0.6
CF4: Agricultural push	−1.3	0.1	0.6	0.6
CF5: India's CS growth path	−1.6	0.7	−2.2	3.1
Panel E: India				
CF1: CS productivity growth	−3.0	0.7	−2.4	4.7
CF2: CS income effect	−3.0	−0.7	6.9	−3.2
Substitution effect (CF2−CF1)	0.0	1.4	−9.3	7.9
CF3: OS productivity growth	−2.1	0.6	2.9	−1.4
CF4: Agricultural push	−1.6	0.1	0.6	0.8
CF5: India's CS growth path	0.0	0.0	0.0	0.0
Panel F: Kenya				
CF1: CS productivity growth	7.4	0.6	−10.9	2.8
CF2: CS income effect	7.4	2.0	−16.8	7.4
Substitution effect (CF2−CF1)	0.0	−1.4	5.9	−4.5
CF3: OS productivity growth	3.9	−0.1	−0.3	−3.5
CF4: Agricultural push	−0.2	0.0	0.1	0.1
CF5: India's CS growth path	−13.3	−0.3	14.8	−1.2
Panel G: Morocco				
CF1: CS productivity growth	−0.7	0.1	0.2	0.4
CF2: CS income effect	−0.7	−0.2	1.8	−0.9
Substitution effect (CF2−CF1)	0.0	0.3	−1.6	1.3
CF3: OS productivity growth	−2.2	0.3	1.5	0.4
CF4: Agricultural push	−0.5	0.0	0.2	0.3
CF5: India's CS growth path	−2.2	0.6	−2.5	4.1
Panel H: Mozambique				
CF1: CS productivity growth	−1.8	0.1	0.7	1.0
CF2: CS income effect	−1.8	−0.5	4.2	−1.9
Substitution effect (CF2−CF1)	0.0	0.6	−3.5	2.9
CF3: OS productivity growth	−1.4	0.1	0.8	0.4
CF4: Agricultural push	−25.8	1.9	10.9	13.1
CF5: India's CS growth path	−0.9	0.2	−1.0	1.7

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Table 7 continued

	Agric.	Manuf.	Consumer Services	Other Services
Panel I: Malawi				
CF1: CS productivity growth	−3.4	−0.3	−1.6	−1.6
CF2: CS income effect	3.4	0.5	−7.6	3.6
Substitution effect (CF2−CF1)	0.0	−0.8	6.0	−5.2
CF3: OS productivity growth	0.5	0.0	−0.1	−0.3
CF4: Agricultural push	−28.5	2.2	12.4	13.9
CF5: India's CS growth path	−4.7	0.9	−2.0	5.8
Panel J: Namibia				
CF1: CS productivity growth	−2.0	0.0	2.1	−0.1
CF2: CS income effect	−2.0	−0.4	4.3	−2.0
Substitution effect (CF2−CF1)	0.0	0.4	−2.2	1.9
CF3: OS productivity growth	−0.5	0.0	0.0	0.4
CF4: Agricultural push	−0.3	0.0	0.1	0.2
CF5: India's CS growth path	−3.8	0.3	1.1	2.4
Panel K: Rwanda				
CF1: CS productivity growth	4.4	0.1	−5.2	0.7
CF2: CS income effect	4.4	0.9	−9.8	4.5
Substitution effect (CF2−CF1)	0.0	−0.8	4.6	−3.8
CF3: OS productivity growth	0.4	0.0	0.1	−0.4
CF4: Agricultural push	−10.6	0.9	4.6	5.2
CF5: India's CS growth path	−7.9	0.4	4.9	2.6
Panel L: Senegal				
CF1: CS productivity growth	6.5	0.5	−10.0	3.1
CF2: CS income effect	6.5	1.3	−14.9	7.2
Substitution effect (CF2−CF1)	0.0	−0.8	5.0	−4.1
CF3: OS productivity growth	5.5	0.2	1.3	−7.1
CF4: Agricultural push	−3.4	0.3	1.5	1.6
CF5: India's CS growth path	−14.7	−0.4	17.9	−2.8
Panel M: South Africa				
CF1: CS productivity growth	−1.9	0.1	1.1	0.8
CF2: CS income effect	−1.9	−0.4	4.4	−2.0
Substitution effect (CF2−CF1)	0.0	0.5	−3.3	2.7
CF3: OS productivity growth	−0.4	0.0	0.1	0.3
CF4: Agricultural push	−0.3	0.0	0.1	0.2
CF5: India's CS growth path	−1.9	0.4	−1.4	2.9
Panel N: Zambia				
CF1: CS productivity growth	−0.7	0.2	−0.4	0.9
CF2: CS income effect	−0.7	−0.2	1.7	−0.7
Substitution effect (CF2−CF1)	0.0	0.4	−2.0	1.6
CF3: OS productivity growth	−0.7	0.2	0.9	−0.4
CF4: Agricultural push	−0.4	0.0	0.2	0.2

CF5: India's CS growth path	−0.9	0.6		−2.5		2.9

Notes: Each entry reports the contribution of a given channel to the change in labour share between 1990 and 2018, expressed in percentage points. All entries are computed as the difference between the baseline and the relevant counterfactual. A positive value indicates the channel increased that sector's labour share; a negative value indicates it reduced it. CF5 reports the additional change relative to baseline if the country had adopted India's CS productivity growth path. Rows need not sum to the baseline change as the channels interact through the model's general equilibrium structure. Preferences are held fixed across all countries at $\rho = -0.5$, $\bar{c} = 0.10$, $\bar{o} = 0.20$.

B.4 Decomposing Structural Transformation: What Drives Agricultural Exit?

	Contribution (pp)				% of Observed Decline			
Country	CS Pull	OS Pull	Ag Push	Residual	CS Pull	OS Pull	Ag Push	Residual
BWA	−1.7	−1.0	0.6	0.1	84.2	49.5	−29.7	−4.0
CMR	0.8	4.0	−4.3	0.0	174.5	872.5	−938.0	−9.1
EGY	−0.1	−1.9	−0.1	−0.1	4.6	87.6	4.6	3.2
GHA	−1.5	−0.7	−1.3	0.0	43.2	20.2	37.5	−0.9
IND	−3.0	−2.1	−1.6	0.1	45.2	31.6	24.1	−0.9
KEN	7.4	3.9	−0.2	0.0	66.8	35.2	−1.8	−0.2
MAR	−0.7	−2.2	−0.5	0.0	20.8	65.2	14.8	−0.8
MOZ	−1.8	−1.4	−25.8	0.0	6.2	4.8	89.0	0.0
MWI	3.4	0.5	−28.5	−0.1	−13.8	−2.0	115.5	0.3
NAM	−2.0	−0.5	−0.3	0.1	73.5	18.4	11.0	−2.9
RWA	4.4	0.4	−10.6	0.0	−75.4	−6.9	181.6	0.6
SEN	6.5	5.5	−3.4	0.0	75.8	64.1	−39.6	−0.2
ZAF	−1.9	−0.4	−0.3	0.0	71.8	15.1	11.3	1.7
ZMB	−0.7	−0.7	−0.4	0.0	38.8	38.8	22.2	0.1

Notes: The CS Pull column reports the contribution of CS productivity growth to the change in agricultural labour share, computed as the difference between the baseline and CF1. The OS Pull column reports the equivalent contribution from CF3. The Agricultural Push column reports the contribution of agricultural productivity growth from CF4. The Residual is the portion of the observed agricultural labour share change not explained by any of the three channels. All contributions are in percentage points (pp) and as a percentage of the observed agricultural labour share change. Positive CS Pull and OS Pull values for Cameroon, Kenya, Malawi, Rwanda, and Senegal indicate these productivity channels were counterintuitively increasing the agricultural labour share — i.e. working against structural transformation. Preferences are held fixed across all countries at $\rho = -0.5$, $\bar{c} = 0.10$, $\bar{o} = 0.20$.

B.5 CF6: Effect of Manufacturing Productivity Growth on Within-Non-Agricultural Reallocation

	Agriculture	Manufacturing	Consumer Services	Other Services
MAR	0.00	−2.13	1.02	1.11
MOZ	0.00	−2.34	1.06	1.28
KEN	0.00	4.07	−2.08	−2.00
MWI	0.00	−0.52	0.24	0.27
GHA	0.00	−0.36	0.18	0.18
BWA	0.00	−2.17	0.87	1.30
EGY	0.00	−1.68	0.69	0.99
SEN	0.00	2.24	−1.07	−1.16
CMR	0.00	0.78	−0.35	−0.43
NAM	0.00	−0.69	0.31	0.38
ZMB	0.00	−1.69	0.78	0.91
IND	0.00	−3.61	1.55	2.06
ZAF	0.00	−1.05	0.45	0.60
RWA	0.00	1.27	−0.59	−0.68

Notes: Each cell reports the change in labour share (percentage points) attributable to manufacturing productivity growth, computed as the difference between the baseline simulation and a counterfactual in which A_m is fixed at its 1990 level. The zero entries for agriculture follow from the model structure: A_m does not enter the agricultural labour share equation, as the Stone-Geary preference specification includes no manufacturing subsistence parameter.

B.6 Regression Analysis

The reformed-form panel regressions (reported in our Discussion section) take the following form, with the full table reported below this:

$$g_{it} = \beta_1 \Delta \ln(CP_{it}) + \beta_2 CS_{it} + \beta_3 I_{it} + \beta_4 HC_{it} + \mathbf{X}'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{18}$$

where g_{it} is annual GDP growth measured as the log difference of real GDP per capita, $\Delta \ln(CP_{it})$ is consumer services productivity growth, CS_{it} is the consumer services employment share, I_{it} is the investment share of GDP, and HC_{it} is the human capital index. The vector $\mathbf{X}_{it}$ contains urbanization controls (urbanization rate, urban primacy, and log population density) included in selected specifications. α_i and λ_t denote country and year fixed effects. The third specification replaces two-way fixed effects with country fixed effects only.

B.7 Regression Table

Table 10: Consumer-Service Productivity Growth and GDP Growth

	(1) Two-Way FE	(2) Urban Controls	(3) Country FE
CS productivity growth	0.095**	0.088**	0.092**
	(0.023)	(0.027)	(0.029)
CS employment share	0.225	0.323†	0.322
	(0.154)	(0.155)	(0.187)
Human capital	−4.227	−0.600	−0.378
	(4.336)	(3.940)	(2.351)
CS productivity level	2.246	4.305	8.392†
	(4.655)	(3.408)	(3.859)
Urban share		−0.151	−0.111
		(0.119)	(0.091)
Urban primacy		−0.046	−0.051
		(0.081)	(0.076)
Log population density		0.380	0.035
		(4.792)	(4.674)
Country fixed effects	✓	✓	✓
Year fixed effects	✓	✓	
Urban controls		✓	✓
Observations	364	334	334
Countries	13	12	12
Adj. R^2	0.231	0.293	0.229
Within R^2	0.069	0.090	0.148
RMSE	3.090	2.742	2.994

Notes: Dependent variable is annual GDP per capita growth (log difference). Standard errors clustered at the country level are reported in parentheses. Column (1) includes country and year fixed effects. Column (2) additionally controls for urbanization, urban primacy, and population density. Column (3) removes year fixed effects while retaining the full set of controls. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, $^{\dagger}p < 0.15$.

C Additional Figures

C.1 12-Sector Overview of Labour Reallocation

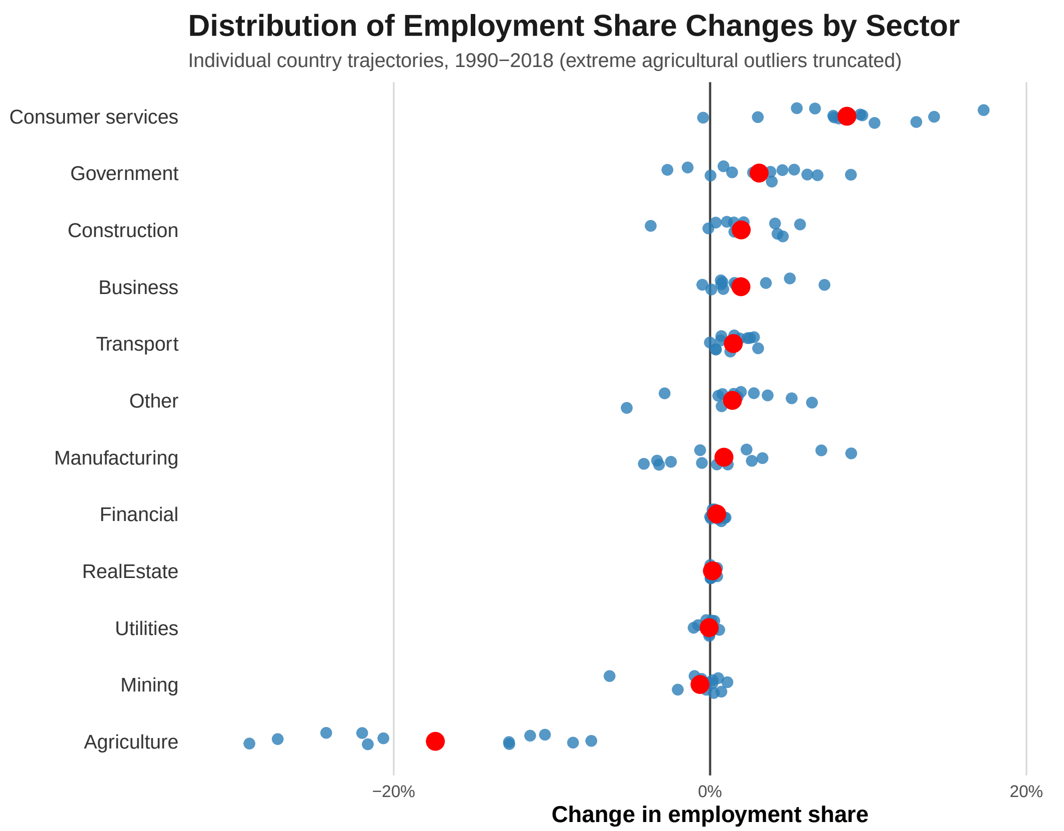

Figure 8: Labour reallocation between 1990 and 2018 across 13 African countries and India. Source: UNU-WIDER ETD. Note: The figure reports changes in sectoral employment shares over the sample period.

D Robustness Checks

We assess the robustness of our results along two dimensions across the 14 countries in our counterfactual sample: the elasticity of substitution ρ, and the Stone-Geary subsistence parameters $\bar{cs}$ and $\bar{os}$. For each dimension we verify that (i) the model remains feasible and well-fitting across the sample, and (ii) the sign and relative magnitude of the income and substitution effects in the counterfactual decomposition are preserved.

D.1 Elasticity of Substitution (ρ)

Table 11 reports model fit under $\rho \in \{-0.25, -0.50, -0.75\}$, with all other parameters held at baseline. The model is fully feasible at $\rho \in \{-0.50, -0.75\}$. At $\rho = -0.25$, Kenya becomes infeasible; all remaining countries are feasible and RMSE values are broadly stable across variants.

Table 11: Robustness Check: Sensitivity of RMSE to ρ

	Country	$\rho = -0.25$	$\rho = -0.50$	$\rho = -0.75$	Feasible?
1	MAR	0.055	0.053	0.057	✓
2	MOZ	0.075	0.072	0.072	✓
3	KEN	—	0.084	0.079	×
4	MWI	0.093	0.089	0.088	✓
5	GHA	0.117	0.102	0.095	✓
6	EGY	0.098	0.103	0.108	✓
7	BWA	0.084	0.103	0.118	✓
8	SEN	0.128	0.106	0.099	✓
9	CMR	0.134	0.123	0.120	✓
10	NAM	0.105	0.124	0.138	✓
11	ZMB	0.130	0.127	0.126	✓
12	IND	0.135	0.132	0.131	✓
13	ZAF	0.143	0.155	0.166	✓
14	RWA	0.158	0.156	0.157	✓

Notes: All structural parameters are held at baseline values: $\alpha_a = 0.30$, $\alpha_m = 0.04$, $\alpha_{cs} = 0.35$, $\alpha_{os} = 0.61$, $\bar{cs} = 0.10$, $\bar{os} = 0.20$. Only ρ is varied. RMSE is the root mean squared error of model-predicted versus observed employment shares, computed over 1990–2018. Countries are ordered by baseline RMSE. The feasibility column indicates whether the country is feasible across all three values of ρ. Kenya is infeasible at $\rho = -0.25$ only.

D.2 Stone-Geary Shifter Terms

Table 12 reports model fit under three jointly-shifted values of $(\bar{cs}, \bar{os})$: the baseline $(0.10, 0.20)$, a downward shift $(0.05, 0.15)$, and an upward shift $(0.11, 0.21)$, with ρ held at -0.50. The model is fully feasible for $\bar{cs} \in [0.05, 0.11]$ and $\bar{os} \in [0.15, 0.21]$ across all 14 countries. RMSE values are stable throughout.

Table 12: Robustness Check: Sensitivity of RMSE to $(\bar{cs}, \bar{os})$

	Country	$\bar{cs}$=0.05, $\bar{os}$=0.15	$\bar{cs}$=0.10, $\bar{os}$=0.20	$\bar{cs}$=0.11, $\bar{os}$=0.21
1	MAR	0.072	0.053	0.049
2	MOZ	0.083	0.072	0.070
3	KEN	0.092	0.084	0.093
4	MWI	0.108	0.089	0.085
5	GHA	0.113	0.102	0.100
6	BWA	0.111	0.103	0.102
7	EGY	0.102	0.103	0.105
8	SEN	0.088	0.106	0.115
9	CMR	0.147	0.123	0.119
10	NAM	0.126	0.124	0.124
11	ZMB	0.134	0.127	0.125
12	IND	0.145	0.132	0.129
13	ZAF	0.148	0.155	0.159
14	RWA	0.190	0.156	0.150

Notes: All structural parameters are held at baseline values: $\alpha_a = 0.30$, $\alpha_m = 0.04$, $\alpha_{cs} = 0.35$, $\alpha_{os} = 0.61$, $\rho = -0.50$. Only $\bar{cs}$ and $\bar{os}$ are varied jointly. RMSE is the root mean squared error of model-predicted versus observed employment shares, computed over 1990–2018. Countries are ordered by baseline RMSE. All 14 countries are feasible across all three variants. Feasibility breaks down at $(\bar{cs}, \bar{os}) = (0.12, 0.22)$, beyond the range reported here.

Across all five parameterizations, the sign of the income and substitution effects is stable for every country in the sample. These groupings correspond precisely to those discussed in Section 6, confirming that our qualitative conclusions are not an artefact of the chosen parameterization.